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MH1200 Linear Algebra I

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1. A Geometric Picture of Subspaces

One intuitive description of a subspace of $\mathbb{R}^n$ is as follows:

It is a k-dimensional hyperplane within $\mathbb{R}^n$ that contains the origin, with $k \leq n$.

Here, the definition of "dimension" will be given later as the maximal size of a basis spanning exactly this subspace.

2. Proving that an Example is a Subspace Rigorously From the Axioms

Example

Example 1

Show that:

$$S = \{(x_1, x_2) \in \mathbb{R}^2 \mid ax_1 + bx_2 = 0\}$$

Is a subspace of $\mathbb{R}^2$.

Solution: First, let us show that S satisfies axiom 1 — i.e., that it is nonempty.

Consider $(x_1, x_2) = (0, 0)$. Notice that:

$$ax_1 + bx_2 = a \cdot 0 + b \cdot 0 = 0$$

Thus, by definition, $(0, 0) \in S$ and thus S is nonempty.

Now, let us show that S satisfies axiom 2 — i.e., it is closed under vector addition.

Suppose $(x_1, x_2), (y_1, y_2) \in S$ are two particular but arbitrarily-chosen elements of S.

By definition, we have:

$$ax_1 + bx_2 = 0$$

$$ay_1 + by_2 = 0$$

Adding both equations, we get:

$$a(x_1 + y_1) + b(x_2 + y_2) = 0$$

Thus, by definition, $(x_1, y_1) + (x_2, y_2) = (x_1 + y_1, x_2 + y_2) \in S$.

Thus, S is closed under vector addition.

Finally, let us show that S satisfies axiom 3 — i.e., it is closed under scalar multiplication.

Suppose $(x_1, y_1) \in S$ were some particular but arbitrarily-chosen element of S and that $c \in \mathbb{R}$ were some particular but arbitrarily-chosen element of $\mathbb{R}$.

By definition, we have:

$$ax_1 + by_1 = 0$$

Multiplying both sides by c, we have:

$$a(cx_1) + b(cy_1) = 0$$

Thus, by definition,

$$c(x_1, y_1) = (cx_1, cy_1) \in S$$

And therefore, S is closed under scalar multiplication, as needed. ■

3. An Equivalent Set of Subspace Axioms

Theorem Alternative Subspace Axioms — Theorem 2

A subset $S \subseteq \mathbb{R}^n$ is a subspace iff:

1. $\vec{0} \in S$
2. S is closed under vector addition
3. S is closed under scalar multiplication

4. Subspace Intersection

Theorem Subspace Intersection — Theorem 3

The intersection of two subspaces S and T is also a subspace.